Graph groups and diagram symmetries

Human Compression research project

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Abstract

The coordinate-kernel theorem for coset geometries applies directly to the binary group attached to a simple graph. We use it to recover the diagram realization of Leemans, Spiga and Tranchida. A residue-counting lemma gives their order bound without a separate rank-three induction base. The isolated minimum among all finite rank-three groups is reduced to three finite orders. These arguments retain the original construction and Frucht's theorem.

1 Sources and common theory

Our principal source is Leemans–Spiga–Tranchida, *Hypertopes with prescribed diagram symmetries*, arXiv:2610.09711v1, nine pages. We recover Theorems 1.1, 3.1–3.3, the displayed group order and the stated rank-three minimum. The final assertion about proper quotients requires preservation of the diagram; without that condition a Boolean quotient is a counterexample. This qualification is not an answer to an open conjecture.

We use the complete common paper *Supported kernels, translation lattices and finite marked quotients*, the companion common paper in this volume, especially UA.SK.Helly and UA.Geometry. The former transfers the common-intersection property of typed cosets through a kernel whose intersections with parabolics are coordinate subproducts. The latter identifies the resulting geometry and its full type-preserving group. These are existing project premises. Their full paper must be charged to a new topic using this application, or once to Topic A when this application is added there. The construction below is the same bilinear construction used in UA.Forest; its coordinate part does not require a forest. No rigidity claim for arbitrary quotients of a cyclic graph is inferred from UA.Forest. We use right cosets throughout, as in that common paper and the source.

2 The graph construction

Let $X = (I, E)$ be a finite simple graph, with an ordering on I. Orient an edge e from its smaller endpoint $t(e)$ to its larger endpoint $h(e)$. On $B_X = \mathbb{F}_2^I \times \mathbb{F}_2^E$ put

$$(v, z)(w, u) = (v + w, z + u + \beta(v, w)), \qquad \beta(v, w)_e = v_{h(e)} w_{t(e)}. \tag{1}$$

Lemma 1. *Equation (1) defines the group with generators r_i, and presentation*

$$r_i^2 = 1, \qquad [[r_i, r_j], r_k] = 1, \qquad [r_i, r_j] = 1 \quad (\{i, j\} \notin E). \tag{2}$$

It has order $2^{|I|+|E|}$, and its subgroup on $J \subseteq I$ is exactly

$$B_J = \{(v, z) : \operatorname{supp}(v) \subseteq J,\ \operatorname{supp}(z) \subseteq E[J]\}. \tag{3}$$

Proof. Bilinearity gives $\beta(v, w) + \beta(v + w, x) = \beta(w, x) + \beta(v, w + x)$, which is associativity. The identity is $(0, 0)$ and the inverse of (v, z) is $(v, z + \beta(v, v))$. Let $r_i = (e_i, 0)$. These are involutions. Their commutator on an edge is its coordinate vector in $\{0\} \times \mathbb{F}_2^E$; other pairs commute. These commutators are

central. Products of the r_i in increasing order produce every $(v, 0)$, and their commutators then produce every central coordinate. Thus (2) maps onto this group. In the abstract presentation its commutators are central. Conjugation by r_i both fixes $[r_i, r_j]$ and inverts it, so its square is one. Collecting the generators in increasing order followed by edge commutators therefore gives at most $2^{|I|+|E|}$ words. The coordinate group attains this bound, proving the presentation and uniqueness. The same collection and the coordinate generators internal to J prove (3) in both directions. The square of (v, z) is $(0, \beta(v, v))$, so every element has fourth power one. If E is nonempty, an edge product has order four and a nontrivial central commutator; the group then has exponent four and nilpotency class exactly two. □

Theorem 2. *The standard coset system of the marking $(r_i)_{i\in I}$ is a regular hypertope Γ_X with full type-preserving group B_X. Its diagram is X with all edges labelled four, and*

$$\mathrm{Cor}(\Gamma_X)/\mathrm{Aut}(\Gamma_X) \cong \mathrm{Aut}(X). \tag{4}$$

Every finite group occurs as this quotient for some connected X.

Proof. Formula (3) gives the intersection property. Project (1) to the vertex coordinates $V = \mathbb{F}_2^I$. The images of the standard parabolics are coordinate hyperplanes. Their typed cosets fix one coordinate each, so they have the common-intersection property. The kernel is the product of the one-dimensional edge groups, and its intersection with B_J retains precisely $E[J]$. The projection is onto every marked subgroup, and V has the intersection property. These verify all hypotheses of UA.SK.Helly. The markers are nonidentity involutions, so UA.Geometry supplies thinness, residual connectedness, regularity, and the asserted full group. The order of $r_i r_j$ is four on an edge, since its square is the nonzero edge coordinate, and is two off an edge. This also identifies every rank-two residue and the diagram.

A graph automorphism preserves the complete presentation (2), hence permutes the markers and the parabolics and induces a correlation of the coset geometry. Conversely, compose any correlation with an element of the regular chamber group so that it fixes the base chamber. Conjugation sends its unique i-adjacency involution to the $\sigma(i)$-adjacency involution, where σ is the type permutation. Thus it preserves pair orders and induces a graph automorphism. If σ is trivial, the normalized correlation fixes a chamber and is type preserving, so UA.Geometry makes it the identity. These constructions are inverse on the quotient in (4). Frucht's theorem supplies a finite connected simple graph with automorphism group any prescribed finite group. If necessary the trivial group is represented by an asymmetric connected tree with more than one vertex. This classical realization theorem is retained, with its original attribution. □

3 Growth in residues

Lemma 3. *Let a finite hypertope of rank $r \geq 2$ have complete diagram, with every rank-two polygon having at least $m \geq 3$ elements of each type. Each type has at least $m + r - 2$ elements. If the hypertope is regular and its chamber group is a two-group, each type has at least 2^r elements, including rank one, and its group order is at least $2^{r(r+1)/2}$.*

Proof. For the first assertion the rank-two base is the hypothesis. Suppose $r > 2$ and fix a type i. Choose another type j and a rank-two residue on i, j. Such a residue exists by completing a flag. Its polygon has at least three elements of each type, so it has a nonincident pair x, y of types i, j. The residue at y has rank $r - 1$ and inherits the same rank-two condition. It contains at least $m + r - 3$ elements of type i by induction. The element x is outside that residue, which proves the asserted strict increase.

For the second assertion start in rank one with its two elements. The same argument gives more than 2^{r-1} elements of each type at rank $r \geq 2$: use an i, j residue with a nonincident pair, and apply the inductive bound to the residue at y. This residue is regular under the stabilizer G_y of y in the original chamber group. Indeed, G_y is transitive on its chambers, and an element in the action kernel fixes a full original chamber and is therefore trivial. Thus its chamber group is a subgroup of the original two-group. Every type size in the whole regular geometry is a subgroup index and hence a power of two. A power of two strictly exceeding 2^{r-1} is at least 2^r.

Choose a chamber and its successive subflags. Orbit–stabilizer identifies successive stabilizer indices with type sizes in residues of ranks $1, \dots, r$. The full chamber stabilizer is trivial, and multiplication of these indices gives the group order. The bounds just proved yield $\prod_{k=1}^{r} 2^k = 2^{r(r+1)/2}$. □

For $X = K_r$, our group has exactly $2^{r+\binom{r}{2}}$ elements, so it attains this bound among two-groups. More generally the same lower bound holds for every regular two-group hypertope with a complete diagram; the argument does not use the common label four. This is a direct extension of the source's counting argument, not a claim of historical priority.

For the source's unrestricted rank-three minimum, take $m = 4$. There are at least five elements of every type, and each corresponding rank-two residue has eight chambers. Thus the group order is at least forty and is divisible by eight. Below sixty-four only orders $40, 48, 56$ remain. The complete program accompanying this application examines all 79 groups of these orders in SmallGroups and all $32,628$ unordered triples of nonidentity involutions. There are 88 triples with all pair products of order four, all in groups of order forty-eight; none generates its group. Every regular rank-three hypertope would supply such a generating triple. Consequently no smaller group occurs. This finite statement depends on the SmallGroups classification and GAP's exact group algorithms. It does not use the source's external hypermap list, and is separate from the uniform written lower bound.

Finally a proper marked quotient retaining the complete all-four diagram cannot be a hypertope, by the two-group bound. Without preservation of that diagram the conclusion fails: $B_{K_r} \to \mathbb{F}_2^r$ kills all edge coordinates. The Boolean maximal-parabolic cosets form the join of r pairs of points, a regular hypertope with empty diagram. The quotient is proper for $r \geq 2$. This gives the exact scope of the closing source assertion.

Attribution and verification

The graph group, its presentation, the diagram realization and the two-group minimum belong to Leemans, Spiga and Tranchida. The source credits Li and Qiao for the construction's inspiration. Here the existing coordinate-kernel argument supplies flag transitivity in one application, and residue growth removes the special induction base. Small-order exclusion is independently recomputed. The companion Lean file verifies the bilinear group law over arbitrary additive abelian groups, its vertex projection, the binary edge form, the support subgroups and their intersection law, and the numerical power-of-two step. It does not identify those subgroups with the generated parabolics or formalize the presentation, geometry, correlation theorem or finite search. No complete formalization, independent review, manuscript admission or measured compression factor is assigned to this working draft.

References

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